Homework 4

Due on May 3

Important. Only handwritten solutions will be accepted; typed solutions are not allowed. In all problems, you must show your reasoning and derivations clearly, step by step. Although it is easy to find answers using online tools, the main purpose of this homework is to deepen your understanding through careful, hands-on problem solving. Working through the details yourself is an essential part of learning the material.

Part I: Integrable Systems

Problem 1 (Kepler's two-body problem)

We consider one of the first classical examples of integrable systems solved by the Liouville theorem: the Kepler two-body problem of planetary motion. Working in the center-of-mass frame, the potential $V(r)$ depends only on the radial coordinate r, and the Hamiltonian is

$$H = \frac{1}{2}\sum_{i=1}^{3} p_i^2 + V(r) \, . \tag{0.1}$$

(1) Show that the angular momentum vector

$$\vec{J} = (J_1, J_2, J_3) \, , \qquad J_{ij} = x_i p_j - x_j p_i = \epsilon_{ijk} J_k \tag{0.2}$$

is conserved, i.e. $\{J_i, H\} = 0$ for each i.

(2) Given the standard symplectic form $\omega = \sum_{i=1}^{3} dp_i \wedge dx_i$, compute the Poisson brackets

$$\{J_i, J_j\} = -\epsilon_{ijk} J_k \, . \tag{0.3}$$

Hence show that the following three quantities are mutually in involution under the Poisson bracket:

$$H \, , \qquad J_3 \, , \qquad J^2 = J_1^2 + J_2^2 + J_3^2 \, . \tag{0.4}$$

(3) Rewrite the Liouville 1-form

$$\alpha = \sum_i p_i \, dx_i = p_r \, dr + p_\theta \, d\theta + p_\phi \, d\phi \tag{0.5}$$

in terms of the spherical polar coordinates

$$x_1 = r \sin\theta \cos\phi \, , \qquad x_2 = r \sin\theta \sin\phi \, , \qquad x_3 = r\cos\theta \, . \tag{0.6}$$

Express the conserved quantities H, J_3, and J^2 in terms of (r, θ, ϕ) and the conjugate momenta (p_r, p_θ, p_ϕ).

(4) Without loss of generality we may rotate coordinates so that $\vec{J} = (0, 0, J_3)$, which amounts to setting $\theta = \frac{\pi}{2}$ (motion in the equatorial plane). Kepler's second law states that the areal velocity is constant; in this setup it is equivalent to conservation of J_3, since

$$\frac{dA}{dt} = \frac{1}{2} r^2 \dot{\phi} = \frac{1}{2} J_3 \, . \tag{0.7}$$

Under this reduction, show that the integral of the Liouville 1-form (0.5) becomes

$$S = \int \alpha = \pm \int^r dr \sqrt{2(H - V) - \frac{J^2}{r^2}} + \int^\phi J_3 \, d\phi \, , \tag{0.8}$$

where the sign $\pm$ is chosen consistently with the sign of p_r.

Derive the equations of motion for the angle variables

$$\psi_H = \frac{\partial S}{\partial H} \, , \qquad \psi_J = \frac{\partial S}{\partial J} \, . \tag{0.9}$$

Discuss their physical interpretation. In particular, under what condition is an orbit closed?

(5) Specialise to the gravitational (Kepler) potential

$$V(r) = -\frac{k}{r} \, . \tag{0.10}$$

(a) **Kepler's first law.** Show that a planet traces an ellipse with the Sun at one focus.

(b) **Kepler's third law.** Let T be the orbital period and a the semi-major axis of the ellipse. Show that

$$T = \frac{2\pi}{\sqrt{k}} a^{3/2} \, . \tag{0.11}$$

(See the Wikipedia page for the relevant terminology on ellipses.)

Part II: Homology

Problem 2

The Euler characteristic of a manifold M is defined in the lecture notes by

$$\chi(M) = \sum_{i \geq 0} (-1)^i \dim H_i(K; \mathbb{R}) \, . \tag{0.12}$$

Given a triangulation $|K| \to M$, show that

$$\chi(M) = \sum_{i \geq 0} (-1)^i \dim C_i(K; \mathbb{R}) \, . \tag{0.13}$$

Problem 3

Show that the Euler characteristic of any odd-dimensional closed orientable manifold is zero.

Problem 4

Find the integer-valued homology groups $H_\ell(\Sigma_g;\mathbb{Z})$ of a closed orientable Riemann surface Σ_g of genus g. Compute its Euler characteristic.

Problem 5

For each of the following non-orientable surfaces, find both the integer-valued homology $H_\ell(M;\mathbb{Z})$ and the real-valued homology $H_\ell(M;\mathbb{R})$, and compute the Euler characteristic:

(a) $M = \mathbb{R}P^2$ (the real projective plane),

(b) $M =$ the Klein bottle.